

Nested Structure in C

Data Hierarchy and Organisation

CodeHive

Problem

🔍 Explore

A **nested structure** is a structure that contains another structure as a member. It is used to represent complex or hierarchical data.

1. Example Definition

```
struct Date {
    int day, month, year;
};

struct Student {
    int roll;
    char name[20];
    struct Date dob; // nested structure
};
```

2. Accessing Members

Use the **dot (.) operator** multiple times to reach the inner members.

```
struct Student s;
s.roll = 1;
s.dob.day = 15;
s.dob.month = 8;
s.dob.year = 2004;
```

3. Initialization

Use nested curly braces for clean initialization.

```
struct Student s = {1, "Avni", {15, 8, 2004}};
```

Why Use Nested Structures? ★

- Organises related data clearly (e.g., DOB inside Student).
- Models real-world entities accurately.
- Improves readability and maintainability of code.
- Enables easy data grouping for complex systems.

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